Standard Operating Procedure for Resolving Routing Protocol Flap (BGP Route Churn, OSPF Adjacency Loss, ISIS LSP Flooding) on Arista Access Node WFD-ACC-71 at Whitefield POP

Node: WFD-ACC-71 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_6776

Description

The Routing Protocol Flap failure domain on the Arista Access Node WFD-ACC-71 at Whitefield POP is attributed to the dynamic nature of BGP, OSPF, and ISIS routing protocols. When a node experiences a high rate of route updates, adjacency losses, or LSP flooding, it can lead to protocol flapping, causing instability in the network. This phenomenon is exacerbated by the complex interactions between the routing protocols, which can result in oscillating routing decisions and subsequent protocol flapping. In the context of the Arista Access Node, this failure domain is further complicated by the vendor-specific CLI dialect and the specific role of the node in the network topology. The node's role as an access node, handling traffic from multiple end devices, makes it a critical component in the network, and any disruption to its functionality can have far-reaching consequences.

Pre-requisite Checks

- Check: Verify that the Whitefield POP is accessible and reachable via remote hands or other available means., Status: Complete

- Check: Confirm that the redundant link status is operational and not contributing to the failure domain., Status: Complete

- Check: Ensure that the necessary permissions and access credentials are in place for the Arista CLI., Status: Complete

Potential Root Causes

Cause	Probability
BGP Route Churn	High
OSPF Adjacency Loss	Medium
ISIS LSP Flooding	Low

Diagnostic Commands & Expected Outputs

```
Command: show ip bgp summary | include Flaps, Expected Output: Flaps:
, Description: This command isolates the BGP route churn by displaying
the number of flap events on the node.
```

```
Command: show ip ospf neighbor | include Down, Expected Output: Down: , Description: This command identifies OSPF adjacency losses by displaying the number of down neighbors on the node.

Command: show isis database | include Flooding, Expected Output: Flooding: , Description: This command isolates ISIS LSP flooding by displaying the number of flooding events on the node.
```

Resolution Steps

Step 1: Step: Clear BGP Flaps, Command: clear bgp * soft, Description: This step resolves BGP route churn by clearing the BGP session and re-establishing it with the neighbors. This is done using the `clear bgp * soft` command, which clears the BGP session and re-establishes it with the neighbors.

Step 2: Step: Reset OSPF Adjacency, Command: clear ip ospf neighbor * soft, Description: This step resolves OSPF adjacency losses by resetting the OSPF neighbor relationships. This is done using the `clear ip ospf neighbor * soft` command, which resets the OSPF neighbor relationships.

Step 3: Step: Stop ISIS LSP Flooding, Command: clear isis database | include Flooding, Description: This step resolves ISIS LSP flooding by clearing the ISIS database. This is done using the `clear isis database | include Flooding` command, which clears the ISIS database.

Post-Check Verification

```
Command: show ip bgp summary | include Flaps, Expected Output: Flaps: 0, Description: This command verifies that the BGP route churn has been resolved by displaying a zero flap count.

Command: show ip ospf neighbor | include Down, Expected Output: Down: 0, Description: This command verifies that the OSPF adjacency losses have been resolved by displaying a zero down neighbor count.

Command: show isis database | include Flooding, Expected Output: Flooding: 0, Description: This command verifies that the ISIS LSP flooding has been resolved by displaying a zero flooding count.
```

Roll-back Procedures

- Step: Restore BGP Flaps, Command: clear bgp * hard, Description: If the fix fails, this step restores the BGP session to its previous state using the `clear bgp * hard` command.

- Step: Revert OSPF Adjacency, Command: clear ip ospf neighbor * hard, Description: If the fix fails, this step reverts the OSPF neighbor relationships to their previous state using the `clear ip ospf neighbor * hard` command.

- Step: Re-enable ISIS LSP Flooding, Command: clear isis database | include Flooding, Description: If the fix fails, this step re-enables the ISIS LSP flooding by clearing the ISIS database using the `clear isis database | include Flooding` command.